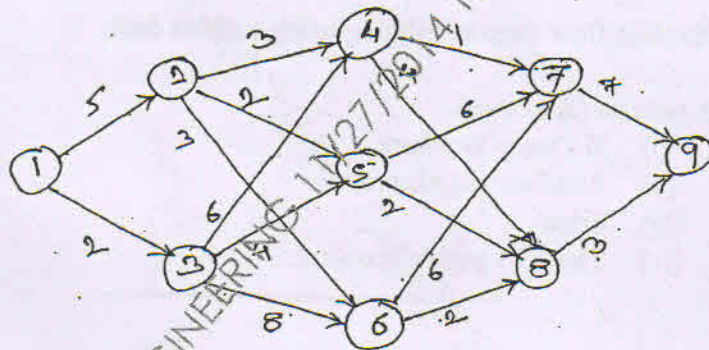


(3 Hours)

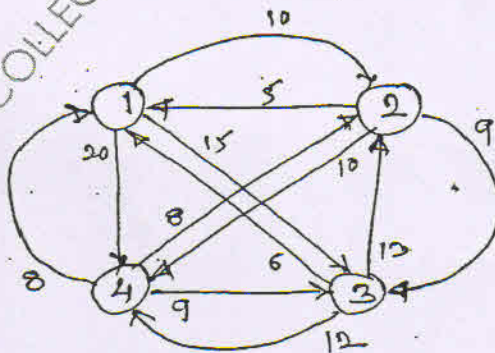
Total Marks : 80

N. B. : (1) Attempt any **four** questions out of **six** question
(2) Assume suitable **data** if **necessary**.

1. (a) Write an algorithm to find minimum and maximum value using divide and conquer and also drive it s complexity. 10
(b) To sort the given set of number using insertion sort and also show the result of each pass. 10
 $\langle 11, 7, 17, 3, 9, 29, 85, 9 \rangle$
2. (a) Find an optimal solution to the knapsack instance $n = 7, m = 15$, 10
Profit = $\{10, 5, 15, 7, 6, 18, 3\}$
Weight = $\{2, 3, 5, 7, 1, 4, 1\}$
(b) Explain optimal storage on tape with example. 10
3. (a) Find a minimum cost path from 1 to 9 in the given graph using dynamic programming. 10



- (b) Find the path of travelling sales person problem of given graph. 10



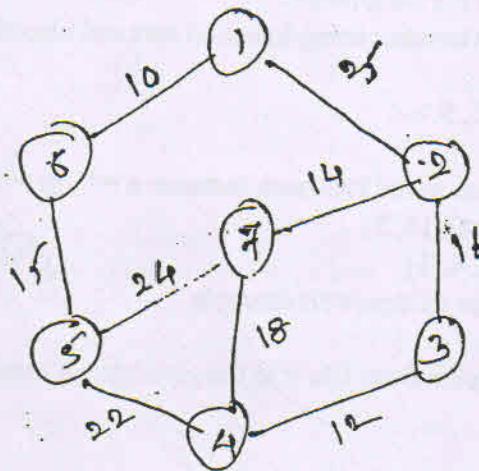
[TURN OVER

BOA/27-11-14 (2/2)

- 2 -

QP Code :12446

4. (a) To generate the Huffman code for given set of frequencies. 10
1, 1, 2, 3, 4, 8, 13, 21
(b) To implement the knuth - Morris-Pratt, string matching algorithm. 10
5. (a) To find MST of following graph using prim's and kruskal's Algorithm. 10



- (b) Explain flow shop scheduling using suitable data. 10
6. Write note on (Any two) 20
(i) N-Queen Problem
(ii) Randomized Algorithm
(iii) Tries
(iv) The 15 - puzzle problem.

GN-Conf:8575-14.